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পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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SUB-SOIL INVESTIGATION REPORT

FOR PROPOSED 10 STORIED (WITH BASEMENT) COM/RESIDENTIAL BUILDING CONSTRUCTION AT GUTATIKOR, SYLHET.
DAG # 1052, 1055, KHA # 3724, J.L # 110, MOUJA- GUTATIKOR.

CLIENT

ABDUL MUSABBIR AHMED

Verification Code: C40W09vp
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH, TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTED BY

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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-80 ft	16
02	Disturbed sample	02	0-75 ft	15
03	Disturbed sample	03	0-80 ft	16
04	Disturbed sample	04	0-80 ft	16
05	Disturbed sample	05	0-80 ft	16
06	Disturbed sample	06	0-80 ft	16
07	Disturbed sample	07	0-80 ft	16
08	Disturbed sample	08	0-80 ft	16
09	Disturbed sample	09	0-80 ft	16
10	Disturbed sample	10	0-80 ft	16

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N;$ $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9 and Table 10** respectively.

Table 7: The allowable skin friction for Precast piles in Tsf (FS=2.5)

Depth	**Bore Hole**									
ft	**BH01**	**BH02**	**BH03**	**BH04**	**BH05**	**BH06**	**BH07**	**BH08**	**BH09**	**BH10**
5	0.074	0.096	0.074	0.074	0.049	0.074	0.074	0.074	0.074	0.074
10	0.096	0.096	0.096	0.118	0.096	0.074	0.246	0.096	0.049	0.118
15	0.096	0.049	0.049	0.049	0.096	0.049	0.096	0.074	0.096	0.096
20	0.118	0.118	0.096	0.074	0.074	0.074	0.096	0.074	0.118	0.074
25	0.074	0.074	0.096	0.118	0.118	0.096	0.118	0.096	0.096	0.074
30	0.174	0.174	0.174	0.218	0.074	0.174	0.118	0.138	0.074	0.261
35	0.074	0.118	0.118	0.096	0.118	0.118	0.118	0.096	0.118	0.096
40	0.096	0.096	0.074	0.118	0.138	0.174	0.174	0.156	0.138	0.118
45	0.118	0.096	0.214	0.096	0.214	0.214	0.218	0.241	0.218	0.196
50	0.375	0.230	0.294	0.138	0.375	0.388	0.428	0.468	0.401	0.482
55	0.401	0.415	0.468	0.268	0.468	0.468	0.468	0.495	0.535	0.508
60	0.669	0.184	0.535	0.669	0.669	0.535	0.669	0.669	0.669	0.669
65	0.168	0.184	0.184	0.669	0.669	0.669	0.669	0.669	0.669	0.669
70	0.176	0.176	0.176	0.669	0.176	0.669	0.176	0.669	0.669	0.669
75	0.168	0.168	0.168	0.168	0.168	0.168	0.168	0.168	0.168	0.168
80	0.168	-	0.168	0.168	0.168	0.232	0.168	0.168	0.168	0.168
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 8: The allowable end bearing for Precast piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.675	0.900	0.675	0.675	0.450	0.675	0.675	0.675	0.675	0.675
10	0.900	0.900	0.900	1.125	0.900	0.675	3.375	0.900	0.450	1.125
15	0.900	0.450	0.450	0.450	0.900	0.450	0.900	0.675	0.900	0.900
20	1.125	1.125	0.900	0.675	0.675	0.675	0.900	0.675	1.125	0.675
25	0.675	0.675	0.900	1.125	1.125	0.900	1.125	0.900	0.900	0.675
30	1.800	1.800	1.800	2.250	0.675	1.800	1.125	1.350	0.675	2.700
35	0.675	1.125	1.125	0.900	1.125	1.125	1.125	0.900	1.125	0.900
40	0.900	0.900	0.675	1.125	1.350	1.800	1.800	1.575	1.350	1.125
45	1.125	0.900	2.925	0.900	2.925	2.925	2.250	4.050	2.250	2.025
50	6.300	3.150	4.950	1.350	6.300	6.525	7.200	7.875	6.750	8.100
55	6.750	6.975	7.875	4.500	7.875	7.875	7.875	8.325	9.000	8.550
60	11.250	36.800	9.000	11.250	11.250	9.000	11.250	11.250	11.250	11.250
65	33.600	36.800	36.800	11.250	11.250	11.250	11.250	11.250	11.250	11.250
70	35.200	35.200	35.200	11.250	35.200	11.250	35.200	11.250	11.250	11.250
75	33.600	33.600	33.600	33.600	33.600	33.600	33.600	33.600	33.600	33.600
80	33.600	-	33.600	33.600	33.600	46.400	33.600	33.600	33.600	33.600
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.048	0.062	0.048	0.048	0.032	0.048	0.048	0.048	0.048	0.048
10	0.062	0.062	0.062	0.077	0.062	0.048	0.160	0.062	0.032	0.077
15	0.062	0.032	0.032	0.032	0.062	0.032	0.062	0.048	0.062	0.062
20	0.077	0.077	0.062	0.048	0.048	0.048	0.062	0.048	0.077	0.048
25	0.048	0.048	0.062	0.077	0.077	0.062	0.077	0.062	0.062	0.048
30	0.113	0.113	0.113	0.141	0.048	0.113	0.077	0.090	0.048	0.170
35	0.048	0.077	0.077	0.062	0.077	0.077	0.077	0.062	0.077	0.062
40	0.062	0.062	0.048	0.077	0.090	0.113	0.113	0.101	0.090	0.077
45	0.077	0.062	0.139	0.062	0.139	0.139	0.141	0.156	0.141	0.127
50	0.243	0.149	0.191	0.090	0.243	0.252	0.278	0.304	0.261	0.313
55	0.261	0.270	0.304	0.174	0.304	0.304	0.304	0.322	0.348	0.330
60	0.435	0.120	0.348	0.435	0.435	0.348	0.435	0.435	0.435	0.435
65	0.109	0.120	0.120	0.435	0.435	0.435	0.435	0.435	0.435	0.435
70	0.114	0.114	0.114	0.435	0.114	0.435	0.114	0.435	0.435	0.435
75	0.109	0.109	0.109	0.109	0.109	0.109	0.109	0.109	0.109	0.109
80	0.109	-	0.109	0.109	0.109	0.151	0.109	0.109	0.109	0.109
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.304	0.405	0.304	0.304	0.203	0.304	0.304	0.304	0.304	0.304
10	0.405	0.405	0.405	0.506	0.405	0.304	1.519	0.405	0.203	0.506
15	0.405	0.203	0.203	0.203	0.405	0.203	0.405	0.304	0.405	0.405
20	0.506	0.506	0.405	0.304	0.304	0.304	0.405	0.304	0.506	0.304
25	0.304	0.304	0.405	0.506	0.506	0.405	0.506	0.405	0.405	0.304
30	0.810	0.810	0.810	1.013	0.304	0.810	0.506	0.608	0.304	1.215
35	0.304	0.506	0.506	0.405	0.506	0.506	0.506	0.405	0.506	0.405
40	0.405	0.405	0.304	0.506	0.608	0.810	0.810	0.709	0.608	0.506
45	0.506	0.405	1.316	0.405	1.316	1.316	1.013	1.823	1.013	0.911
50	2.835	1.418	2.228	0.608	2.835	2.936	3.240	3.544	3.038	3.645
55	3.038	3.139	3.544	2.025	3.544	3.544	3.544	3.746	4.050	3.848
60	5.063	16.560	4.050	5.063	5.063	4.050	5.063	5.063	5.063	5.063
65	15.120	16.560	16.560	5.063	5.063	5.063	5.063	5.063	5.063	5.063
70	15.840	15.840	15.840	5.063	15.840	5.063	15.840	5.063	5.063	5.063
75	15.120	15.120	15.120	15.120	15.120	15.120	15.120	15.120	15.120	15.120
80	15.120	-	15.120	15.120	15.120	20.880	15.120	15.120	15.120	15.120
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed ten storied (with basement) com/residential building.

Soil profile scenario

Overall soil information of the investigated site is irregular in between the boreholes and vertical inconsistency exists in the soil profile.

Soft clay soil layer (grey) with low bearing capacity has been identified at the top regardless of the borehole number. More importantly, there is the existence of black organic soil at every borehole, which in turn makes the investigation site vulnerable. The layer of black organic soil has been identified at 30 ft depth at borehole 07 and at 35 ft depth at borehole 09, but the same layer has been found at 25 ft depth for the remaining boreholes. While a 5 ft thickness of the black organic layer has been noticed at most of the boreholes, the thickness of the said layer has been found as high as 30 ft (between 25-55 ft depth) at borehole 04. Though the soil bearing capacity has been found inadequate up to the depth of 45 ft (regardless of the borehole number), grey hard clay layer with moderate bearing capacity has been observed at 55 ft depth from EGL and the soil bearing capacity is found to be improving at further depth. Irrespective of the borehole locations, grey hard clay/fine sand with good bearing capacity has been observed at 65 ft depth and the scenario continues up to 80 ft depth (75 ft for borehole 02) from EGL. It is to be noted that the client provided data up to the above-mentioned depth for the subsoil investigation of the study area.

From the above discussion, it is very clear that the whole scenario of the proposed site is variable in nature.

Foundation options

The deep foundation (bored/precast pile) may be selected to ensure the stability of the structure and to overcome the ground improvement complication. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found from the attached table in the report (**table 7-10**) which is based on existing soil conditions.

Safety provisions

Since the proposed site is composed of alternate layers of soft cohesive and granular soil, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the **settlement** of the site considering building load while designing deep foundation. Obviously **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile. Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments

The layout of the boreholes of the investigated site, bore-log profiles with SPT values and direct shear test results are shown in the Appendix.

Special notes

These suggestions are absolutely tentative and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that

the above recommendation is restricted only to the existing soil condition of **75-80** ft as client provided.

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